CheckList Set 1

Gamified Virtual Reality Learning Platform for the Sri Lankan School Curriculum with History

(Science Component)

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Executive Summary

The Science component of **Lakmaga** – a gamified virtual reality learning platform for Sri Lanka's O/L (Ordinary Level) curriculum – is evaluated in this feasibility study. This component addresses a significant educational problem: traditional science teaching relies heavily on rote learning with limited lab work, leading to abstract understanding and low student engagement. The proposed solution is an immersive **VR Science module** that allows students to perform virtual experiments and interactive learning quests aligned with their syllabus, making science education more engaging and practical. Key features already prototyped include a virtual paddy field ecosystem, pest control simulation, food chain visualization, human impact scenarios, and biosphere exploration, all designed to enhance understanding through immersion and gamification.

This study examines the project's **technical, operational, market, and financial feasibility**. The findings indicate that the project is **technically viable**, using Unity and standalone VR headsets to create high-fidelity interactive environments (demonstrated by a working prototype of core features). **Operationally**, it can be integrated into schools with proper training and resources, leveraging bilingual content to fit the local context. The **market feasibility** is strong given the lack of localized educational VR content in Sri Lanka[1] and the large number of O/L science students annually[2], though adoption will depend on support for infrastructure and teacher readiness. The **financial analysis** suggests moderate initial costs (for development and VR hardware) that are justified by the potential educational benefits such as improved student engagement and learning outcomes[3].

Overall, the Science VR component is **feasible** and has the potential to significantly improve science learning in Sri Lankan schools. It is recommended to proceed with development and pilot deployment, while addressing key risks (such as ensuring hardware availability, teacher training, and curriculum alignment). With these measures and planned feature enhancements, Lakmaga's science component can become a sustainable and impactful educational innovation.

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Introduction

Background: *Lakmaga* is a collaborative project aimed at enhancing secondary education in Sri Lanka through a gamified virtual reality platform. It was conceived to address gaps in the traditional O/L curriculum delivery – particularly the lack of interactivity and real-world experimentation in science classes. The platform's Science component (the focus of this study) allows students to learn science by exploring immersive ***Ancient Sri Lankan*** virtual environments and conducting simulated experiments. This initiative builds on the rising global trend of using VR in education to improve engagement and learning outcomes[4]. In Sri Lanka, where many schools have limited laboratory facilities and rely on textbook-centric teaching, a VR-based solution offers an innovative way to provide practical, hands-on learning experiences aligned with the national curriculum. The origin of this project lies in the recognition of low student engagement and performance in science at the O/L level, and the opportunity to leverage modern technology to make learning more effective and enjoyable.

Objective: The objective of this feasibility study is to critically evaluate the viability of implementing the *Lakmaga* Science VR module as an individual research component. This involves assessing whether the proposed solution can be successfully developed and integrated from multiple perspectives: technical implementation, operational deployment in schools, market demand/impact, and financial sustainability. The study aims to determine if the Science component is feasible as a standalone module that achieves its educational goals, and to identify any conditions or recommendations for its successful execution. **Key questions include**: Can the required VR technology and content be developed with available tools and expertise? Will schools, teachers, and students be able to adopt and use this solution effectively? Is there sufficient demand and support for a VR-based science learning tool in the target educational context? And what are the cost implications versus the expected benefits?

Scope: This report focuses on the **feasibility aspects** of the Science component of the *Lakmaga* VR learning platform. It evaluates four main areas: **Technical feasibility** (the technical requirements, development process, and challenges for building the VR science module), **Operational feasibility** (the practical implementation in the school environment, including resource needs and usability), **Market feasibility** (the need for and acceptance of the product among stakeholders – students, educators, schools, and alignment with curriculum trends), and **Financial feasibility** (cost estimates for development and deployment, potential funding and returns in terms of educational value). Other aspects such as environmental or ethical feasibility are not the primary focus, since the project's impacts are mainly educational; however, any relevant ethical considerations (e.g. student safety in VR) will be noted. The study is limited to the Science subject component and does not extensively cover the other subject modules of the platform, except as context. It provides evidence-based analysis where possible and concludes with overall feasibility determination, key risks, and recommendations for next steps.

Problem Statement

In Sri Lanka's secondary education system, students often face difficulties in fully grasping science concepts due to traditional teaching methods. The **problem** is that science education at the O/L level is largely theoretical and exam-oriented, with **minimal practical experience** for students. Classrooms rely on lectures, textbooks, and rote memorization rather than interactive or experiment-based learning. Many schools, especially in rural or under-resourced areas, lack well-equipped science laboratories and trained personnel to conduct frequent hands-on experiments. This leads to abstract understanding of scientific concepts and low engagement – students struggle to relate what they learn to real-world phenomena, and their motivation to study science can dwindle[5]. The result is a learning gap where students may pass exams by memorizing facts, but fail to deeply understand scientific principles or develop scientific thinking skills.

The significance of this problem is reflected in student performance and interest in science. Nationally, while over 340,000 candidates sit for the G.C.E. O/L exam each year[2], a considerable number either achieve only basic passes or fail science, partly due to lack of engagement and conceptual clarity. Traditional pedagogical approaches do not sufficiently cater to diverse learning styles - for instance, visual and kinesthetic learners benefit from interactive and visual experiences, which are currently scarce in the standard curriculum. Moreover, there is a growing recognition that merely improving textbook content or adding more facts is not solving the issue; instead, **new methods** of teaching are needed to spark curiosity and understanding.

Another facet of the problem is the lack of integration between curriculum subjects and Sri Lankan historical context. While many countries are leveraging local culture to enrich education, Sri Lankan classrooms often treat subjects like Science and Mathematics in isolation, disconnected from the country's rich historical and technological heritage. This is a missed opportunity - ancient Sri Lankan innovations in agriculture, irrigation, and sustainable living offer powerful, real-world examples that can enhance conceptual understanding. Unfortunately, such cultural content is rarely visualized or integrated into classroom activities. Most students, particularly outside elite urban schools, have never encountered a virtual or interactive representation of historical systems such as the Bisokotuwa or ancient crop management. Without culturally grounded learning tools, both teachers and students struggle to connect theoretical knowledge with real-life relevance. Additionally, the language barrier persists: science is taught in Sinhala or Englsh using many English terms, which adds difficulty for students trying to grasp abstract ideas.

In summary, the problem that the Lakmaga Science VR module seeks to address is the **lack of immersive, interactive learning experiences** in Sri Lankan O/L science education, which leads to low student engagement, superficial understanding, and missed opportunities for practical learning. By tackling this problem, the project aims to enhance science education outcomes –

improving understanding, retention, and enthusiasm for science among students, and ultimately contributing to better performance in O/L science and a stronger foundation for future STEM studies.

Project Description

The proposed solution is an **individual Science VR module** within the *Lakmaga* gamified learning platform. This module is essentially a **virtual reality application** that enables students to learn key science topics through interactive 3D experiences and game-like challenges. Instead of passively reading or listening, learners put on a VR headset and enter simulated ancient Sri Lankan history environments where they can *explore, experiment, and visualize* scientific concepts firsthand. The VR module is aligned with the Sri Lankan O/L science curriculum, ensuring that the content matches what students need to learn for their exams and beyond (e.g. ecosystems, biology, environmental science topics). The design of the module emphasizes *gamification* – incorporating elements like quests, points, and immediate feedback – to motivate students and make learning fun. It also features **bilingual narration and text** (Sinhala and English) so that students can comfortably follow along in their mother tongue while also getting exposure to English scientific terms[6], which is important in the local context.

Key Features Implemented: Several core features have already been developed as part of the Science component prototype, demonstrating the module's functionality:

- **Paddy Field Interaction:** An immersive virtual paddy field environment where students can observe and engage with the rice cultivation process. They can plant virtual rice, add water or fertilizers, and see how different conditions affect growth. This simulates an agricultural ecosystem, teaching concepts of plant biology, soil science, and agriculture's role in the ecosystem. Students learn by doing – for example, managing water levels to see the impact on the crop – which mirrors a real-life paddy field experiment in a safe virtual setting.

- **Pest Control Simulation:** A gamified scenario in which a crop (such as the rice in the paddy field) is threatened by pests. Students take on the role of a farmer/scientist who must identify pest insects and apply appropriate control measures (like introducing a natural predator or using a safe pesticide) to protect the crop. Through this simulation, learners grasp ecological balance and integrated pest management strategies. They see consequences of actions in real-time – for instance, using too much pesticide might harm beneficial insects – thereby understanding the delicate balance in ecosystems and the importance of sustainable practices.

- **Food Chain Visualization:** An interactive 3D visualization of a food chain within the paddy field ecosystem (or a similar habitat). Students can explore different trophic levels

– producers (plants), primary consumers (like caterpillars), secondary consumers (like frogs or birds), etc. The module allows them to literally see energy flow and predator-prey relationships: for example, if the frog population decreases, insect populations explode. This feature reinforces understanding of **ecosystems and food webs** by letting students manipulate variables (e.g., “remove” a species) and observe ripple effects, something impossible to witness in a real classroom. It brings abstract textbook diagrams of food chains to life in a dynamic way.

- **Human Impact Scenarios:** A set of simulations demonstrating **human impacts on the environment**. For instance, one scenario might show what happens to the virtual ecosystem if there is pollution (adding chemical waste to the paddy water) or deforestation (removing trees around the field). Students can observe outcomes like water contamination affecting plant growth and fish survival, or soil erosion when vegetation is gone. Another scenario could be climate-related, such as changes in rainfall patterns. By visually experiencing these impacts, learners gain insight into environmental science topics like pollution, conservation, and climate change, and how human actions can disrupt ecosystems. This fosters a practical understanding of otherwise broad concepts like “human impact on the biosphere.”
- **Biosphere Structure Exploration:** This feature allows students to explore the **layers and components of the biosphere** – essentially understanding how living things and their environments are structured from micro to macro levels. It might include interactive models of ecological hierarchy (organism → population → community → ecosystem → biome → biosphere) or even a virtual “planet Earth” view where students can zoom in on different biomes (forest, wetland, etc.) and see their characteristics. Through this, learners appreciate the broader context of the earlier topics: how a single paddy field is part of a larger biome and biosphere. It solidifies concepts of ecology, showing how local ecosystems tie into global environmental systems.

These features have been implemented using the Unity game engine and tested on a Meta Quest VR headset, confirming that they function as intended in an immersive 3D setting. The prototype’s **high-fidelity environments and interactive mechanics** demonstrate the viability of using VR for such science content. For example, the paddy field scene includes realistic 3D models of rice plants, insects, and weather effects, while the simulations respond in real-time to user inputs (e.g., applying a pesticide triggers an immediate change in pest population in the virtual environment). The module also incorporates narrative elements and guided tasks – a virtual “assistant” might give instructions or ask questions as students progress through a scenario, ensuring educational goals are met. This combination of immersion, interaction, and guidance is designed to maximize learning by doing. Figure 1 (Appendix) illustrates a sample screenshot of the Paddy Field Interaction module, showcasing the user’s first-person view of the virtual field with a heads-up display indicating tasks (e.g., "Water the field to see the effect on plant growth").

Future Features (Planned Enhancements): In addition to the above implemented components, several improvements and new features are planned to enhance the Science module's effectiveness and user experience:

- **Gamified Progress Tracking:** A system will be introduced to track student progress and performance throughout the VR activities. This feature will record completed quests, scores (e.g., how effectively pests were managed), and learning milestones. Students might earn badges or points for completing each simulation, and teachers could access a dashboard to monitor each student's progress. This not only motivates learners through gamification but also provides valuable feedback to educators on student understanding and engagement over time.

- **Improved Food Chain Logic:** The food chain simulation will be refined with more complex and realistic ecological interactions. For instance, additional species may be added to create a full *food web* rather than a linear chain, and factors like carrying capacity and seasonal changes might be included. The logic governing population changes will be made more sophisticated (e.g., incorporating reproduction rates, migration, etc.) to provide a deeper and more accurate simulation of ecosystem dynamics. This will enable advanced exploration, such as demonstrating how an invasive species could disrupt the balance.

- **Enhanced Pest Control Mechanics:** Building on the pest control scenario, the module will incorporate a wider range of pest management strategies and more nuanced outcomes. For example, students could choose between organic farming methods versus chemical pesticides, each with pros and cons. The simulation might include economic factors (a limited budget for pest control measures) to mirror real-life decision making. Enhanced AI for pest behavior will make the challenge more engaging – pests might adapt or hide, requiring students to think critically. These mechanics make the learning experience richer and teach practical decision-making skills in environmental science.

- **Character Animations and Improved Graphics:** To increase immersion, animated 3D characters (such as a farmer guide or villagers in the scene) will be added. These characters can interact with the student's avatar or respond to changes in the environment (for example, a farmer character could cheer when the crop is saved from pests). Additionally, all visual elements will be polished – plants waving in the wind, water flowing realistically, day-night cycles – to create a more lifelike experience. Smoother animations and visual effects will not only captivate students but also help them observe details (like the behavior of animals in the food chain) more clearly.

- **In-VR Assessments:** The enhanced module will integrate assessment tools directly into the VR experience. Quizzes or interactive questions will appear at appropriate junctures – for instance, after a simulation, a student might be asked to predict what will happen if a certain change occurs, or to identify the role of a specific organism in the food web. These in-VR assessments can be multiple-choice questions, matching activities, or even practical tasks (e.g., "demonstrate how you would increase the frog population in the food web"). The system will give immediate feedback. This feature ensures that learning outcomes are being met and gives students a chance to reflect on what they've done in the simulation, reinforcing their knowledge.

- **UI Labeling and Guidance:** User interface improvements are planned to make the learning experience more intuitive. Important objects in the VR scenes (plants, animals, tools) will have labels or tooltips that students can trigger, providing names and brief descriptions (in Sinhala and English) – for example, labeling a dragonfly and noting it's a predator of mosquitoes. The interface will also include clear instructions or a virtual "guidebook" the student can pull up if they are unsure what to do next. These labels and guides help ensure that while the experience is exploratory, it remains educational and aligned to learning objectives. They also aid teachers in debriefing sessions; for instance, a teacher can reference the labels students saw in VR when discussing the lesson in class.

The **methodology** behind the development of this module follows an iterative design approach. It started with identifying challenging topics in the O/L science syllabus (such as ecology and environmental science concepts) that could benefit from visualization and interaction. Each feature was then mapped to specific learning outcomes – for example, the Pest Control simulation targets understanding of ecosystem balance and human environmental impact. Prototypes were developed in Unity and tested in small scale to gather feedback on usability and educational value. The development leveraged best practices from existing VR educational tools; notably, the idea of virtual labs like those in Labster informed the design, but *Lakmaga*'s module is uniquely tailored to local curriculum and languages[7][8]. The VR experiences are designed to last about 5–15 minutes each, making them suitable for classroom periods or homework sessions in a school's computer lab or VR lab. A storyboard and flowchart for each scenario (provided in Appendix A) outline how the student progresses through the tasks and what learning points are covered.

In sum, the Science component of *Lakmaga* is an **innovative educational tool** that transforms textbook science lessons into active learning experiences. By anchoring abstract concepts in virtual yet realistic scenarios, it helps students build intuition and interest in science. The module's features, both implemented and planned, showcase a blend of technology and pedagogy – harnessing VR's immersive power to solve a clear educational need. This feasibility study now analyzes how realistic and practical it is to implement this solution on a wider scale, covering technical, operational, market, and financial considerations.

Feasibility Analysis

Technical Feasibility

Technology and Tools: The development of the *Lakmaga* Science VR module relies on **proven technologies** that are readily available and well-supported. The primary development tool is the Unity 3D game engine, which has robust support for VR development and cross-platform deployment. Unity allows the team to create immersive 3D environments, implement interactive gameplay logic, and optimize performance for standalone VR headsets. The target hardware for deployment is the **Meta Quest** series of VR headsets (e.g., Meta Quest 3), which are standalone devices capable of running high-quality VR applications without the need for a PC. These headsets offer sufficient processing power, graphical capability, and user-friendly experiences (inside-out tracking, handheld controllers) suitable for a classroom environment. By choosing Unity and Meta Quest, the project aligns with industry standards; many educational VR applications use this combination, which means ample documentation, community support, and compatibility with future upgrades. According to the project proposal, the module is being *developed using Unity for deployment on the Meta Quest 3 headset*, featuring high-fidelity interactive environments. This indicates that the technical foundation (engine and hardware) has been validated – the prototype runs on the intended device smoothly.

Development Status and Expertise: A working prototype with core features (paddy field, pest control, etc.) is already in place, which serves as proof-of-concept for the technical feasibility. This prototype demonstrates that the team has the **expertise** to implement VR experiences – skills in 3D modeling, programming, VR interaction design, etc., have been applied successfully to create functional modules. The development process follows modern software development practices, likely using version control (Git) and iterative testing. Each feature was built and tested incrementally, which helps catch technical issues early. The fact that multiple features have been integrated (and presumably the module can switch between a food chain scene, a human impact scene, etc.) showcases that the architecture supports scalability and modularity. **Prototypes or experimental results** from this phase indicate positive outcomes: for example, internal tests showed that students can navigate the virtual world and complete tasks with minimal bugs or performance issues. This reduces technical risk, as we're not starting from scratch – much of the groundwork (such as VR locomotion, user interaction mechanics, and basic UI) has been laid out in the prototype.

Technical Requirements: Implementing this project at full scale involves several requirements. On the software side, continued development in Unity will require a range of assets (3D models of plants, animals, etc., and possibly animations) and possibly integration of external libraries (for instance, a physics engine for realistic interactions, or an analytics SDK to track user performance in VR). These are all feasible since Unity supports such extensions and the assets can either be custom-made or obtained from asset libraries. On the hardware side, deploying to schools means acquiring a number of VR headsets. The Meta Quest headsets are cordless and relatively user-friendly, which simplifies deployment (no need for high-end PCs or complicated setup). Each device would need the *Lakmaga* application installed; updates can be pushed via Wi-Fi. Ensuring that enough space is available for safe VR use in a classroom is another consideration (students need a small 2m x 2m area each to move their arms and turn around safely). This is technically straightforward – it just requires planning in the school setting. **Network connectivity** is not strictly required during usage if the content is fully on-device (which is preferred to avoid dependence on internet), but periodic internet access will help for software updates or if any cloud features (like syncing progress) are implemented. Given many schools have limited internet, the design assumes mostly offline functionality for reliability.

Development Timeline: The remaining development work chiefly involves implementing the *future features* outlined (progress tracking, improved simulations, etc.) and polishing the existing ones. An estimated timeline for technical completion might be on the order of a few months for a small team. For instance, integrating progress tracking and in-VR quizzes might take 4–6 weeks of coding and testing. Enhancing the food chain and pest simulations with more complex logic could require another 4–6 weeks, especially to test the balance and ensure the simulations remain understandable to students. Adding new art assets like character animations and improved graphics is an ongoing process – these can be parallelized, with artists working while programmers handle coding tasks. All in all, a timeline of perhaps 3–4 months of focused development could bring the module from its current prototype state to a feature-complete version ready for pilot testing. This is within a feasible range for an academic project (and aligns with the typical timeline of a final-year project). Indeed, the project proposal outlines development phases spanning several months[9][10], suggesting a structured plan for completing the technical build.

Potential Technical Challenges: While the technology is sound, there are some challenges to acknowledge: - *Optimization:* VR applications need to maintain a high frame rate (ideally 72–90 frames per second on Quest) to avoid motion sickness and provide a smooth experience. As more features and complex simulations are added, performance optimization will be crucial. Techniques like level-of-detail for models, optimized shaders, and efficient coding will be needed. The team must ensure that even with multiple animated creatures in the food web or detailed environments, the application runs lag-free on the hardware. - *User Experience Issues:* VR can cause disorientation or motion sickness if not designed carefully. The module should minimize abrupt motions – for example, using teleportation or slow movement for locomotion, and avoiding scenarios that require the user to move too fast. Ensuring the interface is intuitive in VR (e.g., using gaze or simple controller actions to select objects) is another challenge; however, initial testing likely helped refine this. There is also the matter of accommodating users with glasses or those not accustomed to VR – tutorials or an introductory scene can help users get comfortable with VR controls before diving into content. - *Technical Support and Maintenance:*

Post-development, each VR headset and application installation will require maintenance. If a device crashes or the software has a bug, a plan must be in place to fix it (remote updates, or having an IT person at the school who can re-install or troubleshoot basic issues). For the pilot phase, the development team themselves can handle this, but for scaling up, documentation and training might be needed for school IT staff. This is more of an operational issue but stems from technical complexity.

Technical Feasibility Conclusion: Overall, the technical feasibility of the Science VR module is **high**. The required tools and technologies are accessible and have already been successfully employed in the prototype. The project does not depend on any speculative or experimental tech – VR in education has been implemented elsewhere (e.g., *Labster's virtual labs*) indicating that creating interactive science simulations in VR is entirely achievable with current technology[7]. The development team has demonstrated capability by building key features and plans to follow a structured implementation schedule for remaining tasks. Challenges like performance optimization and user comfort are known issues in VR development and can be mitigated through established best practices and thorough testing. In fact, similar studies suggest VR is expected to improve student engagement and outcomes, though it requires addressing technical challenges such as training teachers to use it[4] – which we will cover in operational feasibility. In summary, from a technical standpoint, there are **no insurmountable barriers**: the Science component can be built and run with the intended technology. The existence of a functioning prototype strongly supports the technical viability of proceeding into full-scale development.

Operational Feasibility

Implementation in Schools: Operational feasibility examines whether the Science VR module can be practically deployed and used in the target environment – Sri Lankan secondary schools (initially at O/L level). One of the major considerations is the **availability of resources** such as VR hardware and space. Each school that adopts the module would need a set of VR headsets (for example, a small class set of 5–10 devices that students can use in rotation). Since most schools currently do not have VR equipment, this would involve introducing new hardware into the school system. Fortunately, devices like the Meta Quest are **standalone and portable**, making them easier to manage compared to tethered VR systems. A single classroom or lab can be designated as the "VR lab" where these sessions take place. Students could use the headsets one group at a time, perhaps during science periods or as part of a lab session. The school administration would need to schedule these sessions so that each class gets an opportunity to use the VR module when relevant topics are being taught (for example, when the syllabus covers ecosystems, the VR session can be scheduled that week). This kind of scheduling and resource sharing is common in schools (e.g., computer labs are often shared), so it is operationally feasible with planning.

Teacher and Staff Involvement: A critical factor is whether teachers and staff can effectively integrate and supervise the VR module. Since this is a relatively new tool, **teacher training** is essential. Science teachers will need to be oriented on how the VR module works, what learning outcomes it addresses, and how to operate the equipment. Training sessions can be arranged during the project's pilot phase – the development team could visit the schools, train a few teachers and ICT technicians on how to set up and run the module, and how to handle basic troubleshooting (like controller battery replacement, recalibrating tracking if needed, etc.). Teachers also need guidance on blending the VR experience with their regular teaching. For instance, they might brief students before the VR session on what to look for (e.g., "notice how the food chain changes if frogs disappear") and afterwards hold a discussion or quiz to reinforce what was learned. This kind of integration is crucial for the VR module to be not just a novelty but a pedagogically sound component of the course. **Operational risk** exists if teachers are resistant or not confident with technology. However, anecdotal evidence and initial feedback suggest that many teachers are eager for tools to improve engagement. In fact, Sri Lankan science teachers in a recent study indicated that virtual labs could boost student engagement, but emphasized the need for proper training and infrastructure to use them effectively[11]. Mitigation here is straightforward: include a comprehensive teacher guide and provide ongoing support (a help hotline or quick maintenance service during the pilot).

Student Adoption and Safety: From the students' perspective, using VR in class is likely to be exciting and engaging. They will need a short orientation on how to wear the headset and use controllers safely. It's important to ensure **safety measures**: for example, students should remain seated or stand in a clear area while in VR to avoid bumping into objects or each other. Because the experiences are mostly stationary (interacting with virtual objects in place), and movement if any is via controller input, the risk of physical accidents is low if properly managed. The school should have rules in place like no walking around blindly with the headset on, and ideally have a supervisor watching. Another aspect is **eye strain or motion sickness** – sessions are recommended to be relatively short (10-15 minutes per student at a time) which is within comfortable limits for most people. Students prone to motion sickness can be identified and perhaps given extra breaks or an alternative learning method. The content itself is designed to be age-appropriate and educational, so there are no significant concerns about inappropriate content. Culturally, the scenarios (paddy fields, local context) are familiar and relevant to students, which should increase comfort and interest.

Logistics and Maintenance: Operating the VR module in a school setting will involve some logistical steps: - Charging and storing devices: The Quest headsets have a battery life of a few hours. Schools will need to charge them after use, so charging stations or power strips in the storage cabinet will be needed. The devices should be stored securely (to prevent theft or damage) when not in use – possibly in a locked cabinet in the lab. Each device can be numbered and assigned during sessions to track usage. - Cleaning and hygiene: With multiple students using the same headsets, hygiene is important. Using detachable face masks (foam interfaces) or hygienic covers can allow quick cleaning between uses. Schools can be provided with guidance on sanitizing the equipment (wiping down with alcohol-based cleaners lightly, etc.). Postpandemic, these considerations are quite standard for shared devices. - Technical support: For daily use, an ICT teacher or lab assistant might oversee the VR sessions. If a device malfunctions, ideally there are a few spares or a backup plan (so the class session isn't derailed). For example, if one headset fails, students could pair up or share remaining devices for that session. Over the longer term, maintenance could involve software updates – since the content may be updated by developers occasionally, the school should connect the headsets to Wi-Fi periodically to get updates or new features. The module is largely offline, but analytics or progress tracking data might be synchronized if implemented; this could be done by connecting the device to a computer and uploading data, or via internet if available.

Scalability and Wider Deployment: In terms of scalability, starting with a pilot in one or two schools is operationally feasible (small scale, easier to manage direct support). If successful, scaling to more schools is possible but will require more coordination. The Ministry of Education or educational authorities could be involved to support a broader roll-out. They might consider setting up VR labs in zonal education centers or teacher training colleges initially. A larger deployment would require bulk procurement of headsets, which is an administrative task rather than a technical one. It would also benefit from establishing a training program for teachers nationwide. This is definitely a bigger challenge, but as a feasibility study for the individual component, we note that such scaling, while ambitious, aligns with global trends where educational tech is adopted gradually. The positive side is that the content can be duplicated easily – once developed, software can be installed on any number of headsets at relatively low incremental cost. The main operational scaling cost is hardware and training.

Curriculum Integration: Operational feasibility also depends on how well the VR module fits into the existing curriculum and school timetable. The Science VR component is explicitly mapped to national curriculum topics[12][13], meaning each VR activity corresponds to lessons students are already learning. For example, the food chain VR might be used during the ecology chapter in Grade 10 science. This mapping makes it easier for schools to justify using class time for the VR module – it's not an extra-curricular game, but a teaching tool tied to their syllabus. Teachers can treat it like a lab session or a visual lesson. Since O/L science is usually taught in periods of ~40 minutes, a VR session could take one period or two. It's feasible to replace or supplement one of the theoretical lessons with the VR practical lesson for better insight. There may be concerns about covering the syllabus on time, but if the VR session actually enhances understanding, it could potentially reduce the need for repetitive teaching of difficult concepts. Additionally, by aligning with curriculum objectives, the module ensures that it complements exam preparation rather than distracts from it – for instance, by including in-VR assessments or follow-up questions, it helps reinforce key points that could be tested.

Language and Accessibility: As noted, the module supports bilingual content (Sinhala and English, potentially Tamil could be added if needed). This is a major operational advantage in the local context[6]. Students from Sinhala-medium schools can fully engage with the VR lessons in Sinhala narration, which means they won't be held back by language comprehension issues. At the same time, seeing English terms in labels or hearing them in voiceover with translation can improve their scientific English vocabulary. This dual-language approach removes a barrier and makes the tool usable across diverse school settings (including rural Sinhala medium schools, which often have students less comfortable in English). It also aligns with national educational policy that encourages bilingual proficiency. In terms of accessibility, VR inherently might exclude some students (e.g., those with certain disabilities like severe visual impairment or motion sensitivity). However, the majority of students can participate, and alternative arrangements can be made if needed (like having some output mirrored on a screen for a student who can't wear the headset, so they can still observe).

Operational Feasibility Conclusion: The operational analysis suggests that **implementing the VR science module in schools is practical** with proper preparation. The main requirements are providing hardware, training teachers, scheduling usage, and ensuring maintenance – all of which are manageable. There will be a learning curve as schools adapt to this new technology, but the enthusiasm it generates can help overcome minor hurdles. The fact that this module addresses a real need (engaging experiments in a resource-limited setting) means that schools and teachers have motivation to adopt it. Indeed, teachers have indicated willingness to incorporate such tools if infrastructure and training are available[14]. By starting with pilot programs and success stories, operational kinks can be ironed out (for example, figuring out the optimal number of devices per class, refining the teacher guide, etc.). Sri Lanka's education system has previously integrated ICT (for example, establishing computer labs over the last two decades); integrating VR is a next step that, while novel, follows a similar pattern of innovation. Considering all factors, the **operational feasibility is moderate to high** – feasible in controlled conditions (pilots) and scalable with institutional support. No insuperable operational barriers emerged: potential challenges like teacher unfamiliarity or device management have clear mitigation strategies. Thus, from an operational standpoint, the project can proceed, ensuring that along with technical development, a focus is kept on training, support, and alignment with school workflows.

Market Feasibility

Target Audience and Market Needs: The primary target audience for the *Lakmaga* Science VR module comprises **Sri Lankan O/L students** (approximately ages 15–16, in Grades 10–11), along with their science teachers and the broader education community (schools, education ministry). In a broader sense, the market could extend to middle-school grades (as the module content could be used for Grades 7–9 as well, per the original project scope) and potentially to A/L (Advanced Level) if expanded in difficulty. The need for this solution in the market is pronounced. As detailed in the Problem Statement, there is a clear gap in engaging educational content for science – students lack opportunities for interactive learning, and teachers lack tools to provide such experiences. Traditional tutoring classes and textbook publishers dominate the supplementary education market, but they largely offer more of the same (notes, lectures, etc.), not interactive or practical learning aids. This VR module is a unique offering that directly addresses a pain point: making science education more effective and enjoyable.

Sri Lanka's education sector has been slowly embracing digital learning, but it remains **underserved**. Digital learning resources (educational software, simulations, etc.) are relatively scarce and often not integrated into the formal curriculum[1]. Many existing e-learning tools are generic or from abroad, thus not well-aligned with the local syllabus or language. This indicates a market opportunity for localized educational content. The *Lakmaga* platform is explicitly designed for the local curriculum and bilingual context, which is a strong differentiator. For example, while a student could watch a YouTube video about ecosystems or play a foreign educational game, none of those would be tailor-made for Sri Lankan exam objectives or available in Sinhala/Tamil. The VR module, by being curriculum-aligned and immersive, meets a need that both educators and students have – to bridge theory with practice in a way that's contextually relevant.

The scale of the potential market is considerable: each year roughly **300,000–350,000 students** sit for the O/L examination in Sri Lanka[2]. These students are spread across over 5,000 secondary schools nationwide. If we envision even a fraction of those schools adopting the VR module, that is a sizable user base. Furthermore, the Ministry of Education has objectives to improve STEM education and often pilots innovative projects that can be expanded if successful. There is also a push for improving IT infrastructure in schools, and a VR-based learning tool could align with those modernization goals.

Stakeholder Analysis: Key stakeholders include: - *Students:* as end users, their "demand" is reflected in how engaging and helpful they find the tool. It's expected that students will be enthusiastic because VR offers novelty and fun in learning. Early feedback (if any informal testing was done with a few students) likely shows high engagement – VR is known to captivate learners, and studies have shown it can particularly increase motivation and focus[4]. Students are likely to talk positively about such experiences, which can create grassroots demand (they might request teachers for more VR sessions). - *Teachers:* their buy-in is crucial, as they are the ones to implement it in class. Teachers generally welcome resources that help explain tough concepts or save time – the VR module could be seen as a way to get concepts across that usually require a lot of lecturing. However, teachers will also evaluate it based on curriculum fit and ease of use. Given that the module is curriculum-mapped and comes with support, teachers can see it as directly helping their teaching outcomes (like better exam results due to improved understanding). Importantly, teachers' perspectives from research indicate that they see strong potential in virtual labs to boost engagement, provided training and infrastructure issues are addressed[11]. If those support systems are in place, teachers can become strong advocates for the tool. - *Schools/Administration:* School principals and administrators care about performance metrics (exam pass rates, etc.) and reputation. If VR can improve science pass rates or the number of students getting high grades (by truly understanding material), it's attractive. Additionally, having modern tech like VR can be a prestige factor for a school, showing they are forward-thinking. The cost might be a concern, but if budgets or grants cover equipment, schools would likely be on board to try it. - *Education Authorities (Ministry, Departments):* At a higher level, the education ministry might view this as a pilot for innovative learning. They often have mandates to improve quality of education and reduce disparities. A localized VR platform fits into that by potentially raising engagement in science across various school types, including rural ones which often lag in resources. It also aligns with the push for digital transformation in education. If the pilot evidence shows improved learning outcomes (for instance, better pre/post-test results or more student interest in science subjects), the authorities could endorse and support scaling it. This could open pathways for funding or incorporating the VR module as an official supplementary resource.

Competition and Alternatives: In the Sri Lankan context, **direct competition is minimal** because this is a pioneering effort in VR for local education. However, it's worth examining both global competitors and local substitutes: - *Global VR Education Platforms:* Products like **Labster** provide virtual science labs covering topics in biology, chemistry, etc., and are used internationally by high schools and universities. Labster's simulations have been shown to increase student performance and engagement in science courses[3]. While Labster is a mature product, it is primarily in English and aligned with international curricula (like AP Biology or IB curriculum) rather than the Sri Lankan syllabus[7]. It also requires relatively good computers or VR setups, and is a commercial product priced for wealthier school systems. Similarly, other VR science offerings (e.g., FotonVR, VRLab Academy) exist, but they target generic science content and none incorporate Sri Lankan context or languages[7][8]. These could be considered indirect competitors, but in practice, a typical Sri Lankan school is unlikely to adopt them due to cost, curriculum mismatch, and language barriers. In fact, the *Lakmaga* project specifically identifies that *no existing solution provides an interactive VR science experience designed for Sri Lankan middle school curricula delivered in local languages*[13]. This suggests the project has a **niche largely to itself** in the local market. - *Local Non-VR Solutions:* The alternative for schools might simply be conventional methods or multimedia. For instance, some teachers use animated videos, PowerPoint presentations, or small-scale experiments with low-cost materials to simulate practical learning. There are also CD/DVD or computer-based educational software provided by publishers, but again, these are not VR and often not interactive in the same immersive sense. Augmented Reality (AR) apps or mobile learning games could emerge as alternatives, but currently, such content specifically for the Sri Lankan O/L science is limited or nonexistent. Thus, the main "competition" is the status quo (textbooks and static diagrams). Convincing the market means showing that VR can do things that traditional methods cannot – such as visualize invisible processes, allow safe experimentation, and dramatically increase student engagement. Given the global research that VR boosts engagement and enjoyment in learning[4], *Lakmaga*'s science module has a strong case.

Evidence of Demand: While this is a new concept, there are indicators of demand: - *Student Performance and Interest:* The persistent issue of science subject failures or low grades suggests students need better learning tools. If a school notices that their science grades improve after using the VR module (as some pilot results might demonstrate), demand from other schools could rise by word of mouth. - *Survey/Interviews:* As part of initial project groundwork (if done), interviewing local teachers and students likely revealed enthusiasm for something like this. (The project's objectives mentioned surveying teachers and students to identify tough topics[15] – those topics have been directly addressed by the VR content). If those surveyed expressed that they would welcome interactive digital help for those topics, that's direct evidence of demand. - *Global Trend:* Internationally, the move towards interactive and experiential learning is growing. Virtual reality in classrooms is being explored in many countries, and early successes (e.g., improved engagement, better lab skills) are reported[4]. Sri Lanka's educators are aware of these trends through conferences or online forums. There is likely a latent demand to not fall behind. If a home-grown solution is available, it could gain traction faster than trying to import foreign solutions.

Market Challenges: Despite the clear need and interest, a few challenges in market feasibility need consideration: - *Cost and Funding:* Schools in Sri Lanka generally operate with tight budgets. Even if the software is provided free or cheaply (say, as an outcome of this research), acquiring VR hardware is an expense. For widespread adoption, funding strategies are needed – possibly government grants, corporate social responsibility (CSR) donations, or phased investments by school development societies. Without external support, poorer schools might struggle to afford VR equipment, which could limit market penetration and also raise equity concerns (a risk that only well-funded urban schools adopt it, widening the resource gap). This challenge is closely tied to financial feasibility and will be addressed in that section; however, from a market perspective, it means that while *interest* might be high, actual *adoption* could be throttled by cost unless mitigated. - *Infrastructure Differences:* As noted, urban vs rural disparity in tech infrastructure is a reality[16]. In rural areas, even consistent electricity or secure device storage can be issues. The market strategy might need to start with more equipped schools and then expand as infrastructure improves (or come with creative solutions like mobile VR labs). - *Cultural and Behavioral Factors:* Some traditional educators might be skeptical of too much "edutainment", worrying that games could distract from syllabus completion. Overcoming this requires demonstrating learning benefits clearly. Also, parents might need awareness; however, seeing their children excited about learning science could win them over. Aligning the VR experiences with exam success (for instance, showing that concepts are understood better) will be key to convincing conservative stakeholders.

Unique Selling Proposition: The *Lakmaga* Science VR module's value proposition in the market is its **localization and curriculum alignment combined with cutting-edge engagement**. It offers what no other product currently does: a fully immersive science learning experience tailored for Sri Lankan students, in their language, fitting their exact syllabus, and enriched with local context (like the paddy field scenario that resonates with Sri Lanka's environment). Additionally, the cross-curricular approach (embedding science in historical settings, as the broader project does) is something novel that sets it apart[8]. Even within the science domain alone, the product can claim that it provides a safe virtual lab where students can do experiments that schools might never be able to afford or risk in reality (like testing effects of pollutants, etc.). This not only fills an educational gap but potentially does so in a cost-effective way – one VR setup can serve many classes and topics, unlike a physical lab that requires maintenance and consumables.

Market Feasibility Conclusion: The market feasibility for the Science VR component is **promising**. There is a clear alignment between the product's offerings and the needs/gaps in the educational landscape. Interest from users (students and teachers) is likely to be strong because the module makes difficult topics easier to learn and teaching more effective. Competition is minimal in the local context, giving *Lakmaga* a first-mover advantage in Sri Lankan VR educational content. The main hurdles – primarily funding and infrastructure for implementation – are external to the product-market fit, and with strategic partnerships (e.g., Education Ministry backing or pilot funding by educational non-profits) these can be overcome. As long as the module can demonstrate measurable benefits (e.g., higher engagement, better test scores in pilot groups), the case for adoption becomes very compelling. In essence, the market is ready for an innovation that modernizes science education, and *Lakmaga* is positioned to deliver just that. Thus, from a market standpoint, the feasibility is favorable: the demand exists, the differentiation is strong, and the potential impact gives a persuasive rationale for stakeholders to embrace the solution.

Financial Feasibility

Cost Estimates – Development: The development of the Science VR module entails costs in terms of time, labor, and technology. As an individual research component (likely developed by a student or a small team), the direct monetary cost has been kept low – leveraging existing hardware and software licenses available at the institution. Unity, for example, has free licenses for students and educational use. The main development "cost" is the effort put in by the developer(s). If we were to value this in financial terms, a commercial development of such a module might require a small team of VR developers, 3D artists, and educators working for several months. Roughly estimated, if this were a startup project, development could cost on the order of **$20,000–50,000** (USD) in manpower when accounting for design, programming, and content creation. However, since this is an academic project, those costs are not out-of-pocket expenses – the student developer's time is part of their study, and possibly there's mentorship but not paid development. Therefore, the initial prototype has been achieved with minimal financial outlay (aside from maybe the cost of one VR headset and a capable PC for development, which might be around $1,500 in total).

To bring the module to a polished product ready for deployment, there might be additional costs for content enhancement (maybe purchasing some 3D asset packages or hiring voice actors for narration). These are relatively small – for instance, asset packs in Unity could be a few hundred dollars, and recording bilingual voice-overs might also be done in-house or at low cost given the educational nature. So the development costs are manageable and mostly one-time. Because the platform can be reused, once developed, the **software can be copied with negligible cost** to multiple devices.

Cost Estimates – Implementation: The more significant financial aspect is the cost of implementing the VR solution in schools. The major cost here is **hardware**: - **VR Headsets:** A Meta Quest 3 headset (as of 2025/2026) costs roughly USD 500 (around LKR 160,000+ after import taxes). If a school buys, say, 5 headsets to run small group sessions, that's about $2,500. For 10 headsets, $5,000. It's a substantial upfront cost for a typical school in Sri Lanka. However, this cost is comparable to equipping a science lab with instruments – for perspective, a school might spend a few thousand dollars on lab equipment over years. And unlike some equipment that serves one subject, VR headsets could be used across subjects (since *Lakmaga* platform has multi-subject modules). - **Accessories and Maintenance:** Minor costs include protective cases, replacement parts (extra controller batteries or charging cables), and perhaps extended warranties. These might add a few hundred dollars over time. - **Infrastructure Prep:** Setting up a space (mats on floor, storage) is minimal cost. Training sessions for teachers might incur costs for trainers or travel; but these could be covered by educational authorities as part of professional development, or by the project's outreach budget.

Funding Sources: Given that asking individual public schools to fund this entirely is challenging, likely funding would come from: - *Government/Education Ministry:* If the value is proven, the ministry could allocate budget to pilot this in target schools (e.g., as part of a science education improvement initiative). The ministry might also leverage donor-funded projects (like World Bank or UNESCO grants that often support ICT in education) to cover such costs. - *Corporate Sponsorship or CSR:* Companies (especially in the tech sector or banks) in Sri Lanka sometimes fund educational technology for publicity and CSR purposes. A set of VR headsets to a school can be a PR highlight for a company. The project team could pitch to such companies for donations of equipment. - *School Development Societies and Alumni:* Some schools have strong alumni networks or parent-teacher associations that raise funds for school improvements. If they see this as beneficial, they might fundraise for the VR kit (similar to how some schools have funded smartboards or new computer labs). - *Phased Approach:* If the cost per school is too high to do all at once, one could phase it: maybe start with 1–2 headsets that rotate among students as a demonstration, then increase numbers as results justify it. Even with few devices, students can take turns (though that reduces immediate impact, it's a way to start). Over time, adding a couple of headsets each year is easier on budget.

For the *Lakmaga* project specifically (being an academic project), immediate funding might be limited to what the university or research grants provide (perhaps enough for a pilot deployment in a couple of schools). But for full feasibility, we consider these broader funding avenues.

Return on Investment (ROI) and Benefits: Since this is an educational initiative, ROI isn't measured in direct revenue (it's not a profit-seeking venture at this stage). Instead, ROI can be thought of in terms of **educational outcomes and cost-effectiveness**: - *Educational ROI:* If VR can improve student understanding, we may see better exam results. For example, if currently only 60% of students pass science and with this tool it could increase to 70% passing (hypothetically), that's a significant educational benefit. It might reduce the need for extra classes or tutoring, which are costs borne by families. Also, improving science education can lead to more students pursuing science at A/L and beyond, contributing to a more skilled workforce in STEM fields long-term (which is a societal ROI). - *Cost Savings:* Over the long run, a VR module can simulate many experiments repeatedly with no extra cost per experiment. Traditional labs require chemicals, specimens, equipment that can break, etc., and not all schools can afford those for every student. VR provides those experiences virtually. For instance, dissecting a frog virtually means no actual specimens are needed, and every student can "do" it rather than just watch a teacher do it once. While VR doesn't fully replace physical labs for all cases, it can fill gaps where physical labs are absent. There's potential cost saving in not having to set up certain expensive apparatus for real (e.g., measuring pollution effects or doing ecological field studies would be expensive or impossible practically, but VR can show it). - *Efficiency:* Once a VR lab is in place, it can serve multiple classes and even multiple subjects (if the platform expands). That multiplies its value. For example, the same set of headsets can run a History VR scenario or a Math visualization tool. So the investment spreads across disciplines.

There have been indications from global usage that virtual labs lead to improved grades and retention[3]. One data point from Labster: 82% of students were highly engaged and course pass rates increased, sometimes by a full letter grade, when virtual labs were used[3][17]. If *Lakmaga* yields anything close to that, the educational return is enormous for the cost of a few devices.

Break-Even Considerations: If we were to imagine this as a project that needs justification, the break-even in educational terms would be when the benefits (improved results, reduced dropout in science, etc.) outweigh the costs of hardware and training. For example, if a school spends $5,000 on a VR kit and as a result a certain number of students improve enough to qualify for science-related A/L or careers who otherwise might not have, that could be seen as paying off indirectly. Such benefits are qualitative, but one could quantify by saying: each year, that $5,000 kit is used by 200 students; over 5 years that's 1,000 student-experiences, i.e. $5 per student for an enriching science experience. Compared to costs of tuition classes or lab equipment per student, that can be reasonable.

If considering a commercial angle, one could think of selling this as a package to schools or the ministry. The software development might be funded by a grant, and afterwards maintenance and updates could be a small subscription or support fee. However, since this is an academic component, immediate financial profit is not the goal – sustainability is.

Long-term Financial Sustainability: For the project to sustain after initial rollout, there would need to be some ongoing investment: - Devices might need replacement every few years (VR tech evolves, or wear and tear). - Software updates and possibly extension to more content will require developer time. Perhaps the developers could open-source or license the content to the Ministry, which then takes over maintenance (for instance, NIE – National Institute of Education – could incorporate it into their resources). - Alternatively, the project could spin off as an edu-tech startup seeking impact investment to continue development and support.

However, given the likely support from educational authorities if proven effective, it's plausible that funding for maintenance (like upgrading to new headsets or adding a Tamil version) can be obtained as part of national education budget lines or ICT in education programs.

Financial Risks: There are some financial risks: - If costs of VR hardware don't drop or if they increase, it may be harder to procure many devices. Currently, technology prices often drop over time – by the time of scaling, there might be cheaper VR options or used devices in market. - If the project does not show clear improvements, convincing funders to invest more will be tough. Thus, collecting data on the educational impact is important to justify further financial input (we might recommend doing an ROI study after pilot – e.g., measure improvement in test scores of VR users vs non-users). - Another risk is damage or loss of devices which would mean replacement costs; hence, emphasizing proper handling and providing protective measures is necessary to protect the investment.

Financial Feasibility Conclusion: Financially, the project requires moderate investment but nothing extraordinary for an educational innovation. The **costs are largely one-time and front-loaded** (hardware purchase, content development), whereas the benefits accrue over many years of usage. When scaled to multiple schools, the total costs rise, but so do total beneficiaries. Given the interest from institutions to improve STEM learning, it is reasonable to expect that funding can be sourced for initial pilots (through grants or public funding). If those pilots demonstrate success, the cost to expand would likely be seen as an investment into the country's education quality.

The feasibility in financial terms is thus contingent on securing initial funding support – which is achievable, especially if the project is framed as aligning with national priorities in education. The **value proposition (better learning at a relatively low per-student cost)** helps make the financial case. In a scenario where budgets are constrained, options like starting with fewer devices or targeting specific schools (maybe model schools in each region) could be employed to distribute cost over time. Importantly, because the module can lead to intangible benefits such as increased engagement and possibly improved exam performance, one could argue that the cost per improved outcome is justified.

In summary, **financial feasibility is reasonable**: the project does require funding but not beyond what can be mustered for a promising educational pilot. If managed well with stakeholder support, the financial aspect should not be a blocking issue. Furthermore, the potential for cost-sharing or phased investment improves the outlook. As VR technology becomes more common, costs are likely to decrease, making it even more viable in the long term. Therefore, with a clear plan for funding and demonstrating ROI in educational terms, the Science VR module passes the financial feasibility test.

Risk Analysis

Implementing the *Lakmaga* Science VR component comes with several **risks** across technical, operational, financial, and market dimensions. It is important to identify these risks, assess their likelihood and impact, and propose mitigation strategies for each.

- **Technical Risks:** One technical risk is the possibility of *software bugs or performance issues* that could disrupt the learning experience. For example, the VR application might crash or freeze, or certain interactive elements might not work as intended under heavy usage. The likelihood of minor bugs is moderate (as with any software), but critical failures can be kept low with thorough testing. The impact of a technical failure in class is that the session is interrupted and it could undermine confidence in the tool. **Mitigation:** Perform extensive testing under conditions similar to actual use (testing on the Quest devices, with multiple sessions back-to-back to catch memory leaks or performance degradation). Have a fallback plan in class – for instance, if the VR doesn't work, the teacher could switch to a discussion or a backup demonstration for that day, to avoid lost time. Additionally, ensure developers are on standby during initial pilot rollouts to quickly fix any unexpected bugs. Over time, maintenance updates should be scheduled (e.g., a check every term to apply any patches). Using Unity's established frameworks and adhering to VR development best practices reduces the risk of technical breakdowns significantly.

- **Hardware Risks:** The VR headsets and controllers are physical devices that can malfunction, get damaged, or become obsolete. There's a risk that devices might break (due to being dropped by students, for instance) or batteries might fail. The likelihood of occasional hardware issues is fairly high (devices always face wear and tear), and the impact is the reduction in available equipment, possibly hindering a planned class if not enough units work. **Mitigation:** Proper training on device handling for both teachers and students can minimize accidental damage (e.g., using the headstrap securely, not walking around with the headset on, etc.). Provide protective accessories like silicon covers or tether straps for controllers. Keep a small budget or inventory for spares – e.g., an extra headset or extra controllers in each kit. In terms of obsolescence, plan for upgrades; perhaps after 3-4 years, headsets may need replacement or new models might offer better safety/features – keeping an eye on technology trends and budgeting for periodic upgrades will mitigate the risk of the platform becoming unsupported.

- **User Experience Risks (VR-specific health/safety):** There's a risk that some students might experience *motion sickness, eye strain, or discomfort* using VR. The likelihood is low to moderate (modern VR has improved, but a subset of users – maybe 5-10% – could feel dizzy or nauseous). The impact on those individuals is unpleasant, and on the class it could cause concern about the tool's safety. Also, overuse could cause headaches or fatigue. **Mitigation:** Design the VR experiences with comfort in mind – use teleportation or slow movements, avoid intense motion. Also limit VR session durations (10-15 minutes at a time for beginners). Before using VR, screen students for any known issues (for instance, epilepsy or extreme motion sickness susceptibility) and perhaps have them opt out or use under caution. Always allow students to remove the headset if they feel unwell, and ensure classrooms are ventilated and at comfortable temperature (since headsets can be warm). Including breaks between VR sessions is important. By following these measures, health-related incidents can be kept minimal. It's also good to educate students: e.g., "focus on a fixed point if you start to feel dizzy, or take off the headset slowly."

- **Curriculum and Efficacy Risks:** A risk exists that the VR module might not *align perfectly with curriculum changes* or might not directly translate to better exam performance if not used correctly. Syllabi can change over time; if content isn't updated, it could become less relevant. Also, teachers might fear that time spent in VR might take away from exam preparation. Likelihood of minor misalignment grows over years, and if results don't improve, teachers may lose interest (impacting the project's longevity). **Mitigation:** Maintain close alignment with curriculum by updating the content when syllabus updates occur (the developers or future maintainers should be ready to tweak scenarios or add new topics if needed). Involve curriculum experts during development to ensure all necessary points are covered. As for demonstrating efficacy, incorporate assessments and gather data: for instance, have students take a short quiz after the VR session and share results with teachers – if scores are good, teachers see the value. Conducting a controlled study during pilot (one class uses VR and another doesn't, then comparing their test results) can provide evidence. By proving that VR usage either maintains or improves test performance, this risk can be mitigated. Teachers can then feel confident that it's not just a fun add-on but a truly effective tool.

- **Teacher Adoption and Training Risks:** If teachers do not fully buy-in or feel under-prepared to facilitate VR sessions, the project could falter. This risk is moderate in likelihood – some teachers might be hesitant with new tech or worried it's complicated – and the impact is significant because teachers are gatekeepers to classroom implementation. **Mitigation:** Provide comprehensive training sessions and easy-to-follow guides. Possibly designate a *tech-champion* teacher at each school who is enthusiastic, to lead by example and help peers. The project can start with willing teachers to generate success stories. Showcasing positive feedback from students and maybe having teachers observe a session led by the developers or a trained instructor initially can build confidence. Continued support (like a helpdesk or quick troubleshooting guide) will ease their concerns. Recognize teachers' time constraints – integrate the module with minimal extra work for them by supplying lesson plans or slides to accompany the VR activity. Essentially, make the tool as teacher-friendly as possible. With these steps, the likelihood of teacher resistance drops and any that remains will likely vanish after they see the student enthusiasm and learning gains.

- **Financial and Resource Risks:** On the financial side, as discussed, the risk is that funding might be insufficient to get enough hardware or sustain the project beyond prototypes. The likelihood depends on external factors (budget approvals, etc.), and the impact is the inability to scale or even properly pilot. **Mitigation:** Mitigation includes preparing a strong case with data (to convince stakeholders of the ROI), seeking multiple funding sources (don't rely on just one grant – apply to several or get partial contributions from multiple sponsors), and scaling the project according to available resources (if only 2 headsets are funded in a pilot, use them efficiently in rotation rather than waiting for 10). Also, optimize costs where possible: e.g., use existing school computers for certain parts of training or use older model headsets if they are cheaper but still effective. By demonstrating even a small-scale success, it can unlock further funding ("nothing succeeds like success"). Planning for sustainability – perhaps training local teachers or university students to continue development as part of their projects – can mitigate risk of the project dying after initial creators move on.

- **Operational Disruption Risks:** There's a minor risk that introducing VR could cause *schedule disruptions or management issues* in school. For instance, if many students are eager or if logistic scheduling isn't sorted, it could clash with other classes, or there might be times when devices aren't charged, etc. Likelihood is low with good planning, but impact could be frustration or reduced trust in the tool. **Mitigation:** Work with school timetables carefully – maybe assign VR sessions in a fixed slot (like during science lab periods). Ensure someone is responsible for device readiness (charged and set up). Also manage student expectations – they should see it as part of learning, not an anytime toy. Clear rules and schedules will prevent chaos. Essentially treat the VR module as you would a computer lab session or science lab – structure it.

- **Market Acceptance Risks:** If the project aims to expand, a risk is whether schools beyond the pilot will adopt it, or whether policy-makers support it. This overlaps with market feasibility; reasons for reluctance could be skepticism or competing priorities. The mitigation for this is largely in *communication and demonstration*. Share results in educational forums, invite other teachers to observe pilot classes, and highlight student success stories. Garner endorsement from influential educators or officials by involving them early (perhaps having Ministry observers at pilot trials). By proactively engaging the education community and showcasing positive outcomes, the risk of stagnation is reduced.

In evaluating these risks, none appear to be fatal to the project – they are typical challenges for introducing innovative technology into education. **Most risks are manageable with careful planning and responsive strategy.** For example, technical and hardware issues can be managed with testing and support, user experience issues with good design and guidelines, and adoption issues with training and evidence of success. The highest impact risk – lack of funding or support – can be mitigated by strong stakeholder engagement and phased implementation. It's also worth noting that many of these risks have been encountered in similar projects worldwide; thus, there is prior knowledge on how to handle them (e.g., numerous case studies exist on integrating ICT in classrooms, which provide playbooks for training and maintenance).

In conclusion, the risk analysis reveals that while there are multiple risks to address, **none of them are insurmountable**. With the recommended mitigation strategies in place, the likelihood and impact of these risks can be greatly reduced. This means the project can proceed with a good degree of confidence, so long as risk management is treated as an ongoing part of the project implementation. A risk log will be maintained throughout the project, and contingency plans (like having backup hardware or alternate lesson plans) will be ready to ensure that the educational process is never adversely affected. By anticipating and managing these potential pitfalls, the Science VR component can be rolled out smoothly and successfully.

Recommendations

After conducting the feasibility analyses, this study finds that the Science component of the *Lakmaga* VR learning platform is **viable** and holds significant promise for improving O/L science education in Sri Lanka. The following recommendations are put forward to ensure successful implementation and maximization of the project's impact:

1. Proceed with Phased Implementation: It is recommended to move forward with the project, starting with a well-planned pilot phase. Begin by deploying the Science VR module in a small number of selected schools (for instance, one urban and one rural school) to pilot the approach. This phased implementation will allow the team to gather feedback, demonstrate success, and refine the module in a controlled setting before scaling up. A phased approach also helps manage resources – focusing initial hardware and training investment where it can have the most learning, and then leveraging pilot results to secure additional funding for expansion.

2. Secure Funding and Partnerships: To address financial feasibility, seek partnerships with stakeholders such as the Ministry of Education, corporate sponsors, or international educational grants. Present the pilot plan and the feasibility findings to potential funders, highlighting the innovative nature of the project and its alignment with educational improvement goals. A recommendation is to prepare a cost-benefit brief including data from this study (e.g., number of students impacted per investment, and the evidence that VR can improve engagement/outcomes[4][3]). Aim for a mix of funding sources – for example, a government grant to cover equipment for public schools, and CSR funds from a tech company to cover content enhancement or teacher training workshops. Additionally, partnering with teacher training institutions can embed the VR module training into professional development, ensuring longer-term support.

3. Enhance and Finalize Technical Development: Continue the development of planned features to improve the module's effectiveness. In particular, prioritize the **gamified progress tracking and in-VR assessment** features, as these will provide measurable data on student learning and enable teachers to monitor progress. Implement the **improved food chain and pest control logic** to increase the depth of content, but ensure they remain user-friendly. Before broader deployment, run a comprehensive quality assurance pass on the software, possibly involving external testers or a small group of students, to catch any bugs or usability issues. It's recommended to develop a *"Teacher Mode"* or demo mode in the application which allows an instructor to walkthrough the content quickly – this can be useful for demonstration in teacher training sessions or monitoring what students are seeing on a separate screen.

4. Develop Training and Support Materials: To ensure operational success, prepare detailed yet accessible **training materials** for teachers and school IT staff. This should include a user manual, quick-start guides with illustrations, and troubleshooting FAQs. Moreover, conduct hands-on training workshops before the pilot launch: show teachers how to use the headsets, run the module, and integrate it into their lesson plans. Provide lesson plan suggestions or worksheets that complement the VR sessions; for instance, a worksheet where students note down observations from the VR and answer questions, bridging the VR experience with writing and reflection. It's also recommended to designate a support contact (either an email/phone support or a local tech coordinator) during the pilot phase so that any issues encountered by schools can be resolved swiftly. Building teachers' confidence and competence with the technology is a key recommendation, as it directly influences student experience.

5. Monitor, Evaluate, and Document Outcomes: Establish a mechanism to monitor the pilot implementation and evaluate its outcomes. This could involve pre- and post-intervention surveys for students and teachers (asking about engagement, understanding of topics, etc.), and comparing student performance on relevant test questions before and after using VR. Collecting such data will provide concrete evidence of the module's impact (e.g., improved scores, increased enthusiasm in class, etc.). Additionally, document qualitative feedback – anecdotes or observations such as "students were excited and asked deeper questions after the VR lesson" can be powerful. This evaluation is recommended not only for proving feasibility to stakeholders but also for identifying any areas of improvement in the content or approach. After the first pilot cycle (perhaps one term or one academic unit), convene a review meeting with all stakeholders to discuss findings and make any necessary adjustments to both the module and implementation strategy.

6. Mitigate Identified Risks Proactively: Based on the risk analysis, implement mitigation strategies in advance. For example, procure a few **spare headsets and accessories** so that device failure doesn't derail classes. Set clear guidelines for safe VR use and incorporate them into student orientation sessions. Schedule VR sessions thoughtfully to avoid disrupting core teaching time – perhaps as an enrichment during double periods or as part of lab days. By following the risk mitigation suggestions outlined (training, technical support, health precautions, etc.), many issues can be prevented or minimized. It is recommended to create a risk checklist for the implementation team to use as a reference, ensuring that each risk area (technical, operational, etc.) has an owner and a plan.

7. Engage Stakeholders and Build Awareness: Increase buy-in and enthusiasm by actively engaging with the education community. Invite education officials, school principals, and other science teachers to observe the VR sessions in action during the pilot. If possible, let a few students demonstrate what they learned via VR in a forum (like a science day event or a staff meeting). Showcasing the module's capabilities can generate word-of-mouth support. Simultaneously, keep parents informed – perhaps through a demonstration at a parent meeting or via newsletters – to illustrate how this new tool is benefiting their children's learning. Building a broad base of support will help when expanding the program. It's recommended to also document the project in a short video or report that can be shared publicly, as this could attract further support or partnerships.

8. Plan for Long-Term Integration: Look beyond the pilot and consider how the Science VR module could be sustained and possibly expanded. For instance, if the pilot is successful, recommend integrating the module into the regular science curriculum support materials – the Ministry could include references to the VR activities in teacher guides or future curriculum revisions, solidifying its role. Additionally, plan for integrating any new relevant content (for example, if climate change is given more emphasis in curriculum, ensure the module covers it in the human impact scenarios). From a platform perspective, maintain the flexibility to update or add content, and encourage feedback loops where teachers can suggest improvements or new features. A recommendation is to explore **localization** to Tamil medium as well (if not already), to cover all linguistic groups, especially if scaling nationwide. Long-term integration might also involve expanding the concept to other subjects (since *Lakmaga* is multi-subject), so ensure that the science component can serve as a model for others by keeping thorough documentation of its development and outcomes.

9. Foster a Community of Practice: Encourage the creation of a community or network of teachers and schools using the VR module. This can be as simple as a WhatsApp or online forum where teachers share tips, lesson integration ideas, and troubleshoot together. Such peer support can be incredibly effective and reduce the burden on the core team. It also helps spread best practices – for example, one teacher might discover a good approach to debrief the VR session that others can adopt. The recommendation is for the project team to facilitate this initially (maybe by setting up the group and prompting discussions) and then let it grow organically. In the long run, this community can advocate for the program and help maintain momentum.

10. Overall Feasibility Conclusion: In summary, it is recommended that the Lakmaga Science VR project be **approved and supported for continuation** into implementation. The feasibility study indicates strong potential benefits that outweigh the costs and challenges. The **final recommendation** is to use this study's findings as a roadmap: proceed with development completion, implement in a controlled setting, and use data-driven evaluation to guide wider rollout. By following the above recommendations, the project team can increase the likelihood of a successful outcome – where students gain a richer understanding of science, teachers gain a valuable teaching aid, and the education system gains an innovative model for tech-enhanced learning.

Conclusion

This feasibility study examined the proposed **Science component of the "Lakmaga" Gamified VR Learning Platform** in detail, covering its introduction into Sri Lanka's O/L science curriculum from multiple angles within an ancient Sri Lankan culture. The analysis reveals that the project is not only **feasible** but also timely and potentially transformative for science education.

In the **Introduction**, we outlined the vision: leveraging immersive virtual reality technology to address the longstanding issue of low engagement and limited practical exposure in science classes. The **problem statement** highlighted how traditional teaching methods leave many students disengaged and memorizing rather than truly understanding scientific concepts. The *Lakmaga* Science VR module offers a compelling solution to this problem by bringing science experiments and phenomena to life in a way that is aligned with the national curriculum and accessible in local languages.

Through the **Project Description**, we saw concretely how the VR module works – from virtual paddy fields to food chain simulations – demonstrating an innovative blend of education and technology. These features show how abstract concepts can be made concrete and how students can learn by doing in a safe, controlled virtual environment. It also sets a foundation for future improvements, indicating that the project has room to grow and adapt (such as adding progress tracking and richer interactions).

The core of this report assessed **feasibility aspects**: - **Technical Feasibility:** We found that the project stands on solid technical ground. The use of Unity and Meta Quest VR is appropriate and validated by a working prototype. There are no insurmountable technical barriers – any challenges like performance optimization are routine in VR development and can be handled. Similar VR educational applications exist globally, reinforcing that what is being attempted here is within reach of current technology[7]. - **Operational Feasibility:** Implementing the VR module in schools is practical with good planning. While introducing VR is novel, schools can integrate it akin to a computer lab or science lab activity. Training teachers and managing the hardware will be key, but we outlined clear strategies for these, and educators have expressed willingness to use such tools given support[11]. The bilingual and curriculum-fitting nature of the content ensures it plugs into classrooms relatively seamlessly[6][1]. - **Market Feasibility:** There is a significant need and appetite for this kind of solution. With hundreds of thousands of students and a lack of localized interactive learning resources, *Lakmaga* fills a market gap[1]. We noted that no direct local competition exists and that stakeholders – students, teachers, and education authorities – are likely to support the initiative, especially if early results show improved engagement. The market analysis gave confidence that scaling up is possible if initial success is demonstrated. - **Financial Feasibility:** While there is a cost to acquiring VR hardware and investing in development, these costs are reasonable and can be justified by the benefits. We reasoned that the cost per student reached is low when spread over usage, and avenues for funding (government, CSR, etc.) are viable. The educational returns (improved learning outcomes) provide a strong argument for the investment[3]. Thus, financially, the project can be supported with strategic resource mobilization.

The **Risk Analysis** did not uncover any "showstopper" risks; rather, it enumerated challenges that can be managed through proactive measures. From technical glitches to teacher adoption, each risk has a mitigation plan in place. This indicates that with careful execution, the project can avoid pitfalls that have derailed some tech-in-education efforts in the past.

Bringing all these points together, the study's key finding is that the Science VR component is **feasible and worth pursuing**. The potential impact on student learning is substantial: we expect to see students more excited about science, better able to visualize and understand complex processes, and ultimately performing better academically due to deeper comprehension. In qualitative terms, success would look like a classroom where students are actively discussing what they saw in VR, asking insightful questions ("I saw the frogs disappear and then there were so many insects – is that what happens in real life?"), and teachers are able to build on those experiences to reinforce theory with practical insight. Moreover, this project could serve as a **model for educational innovation** in Sri Lanka – paving the way for more VR or technology-enhanced learning tools in other subjects and grades, thereby contributing to modernizing the education system as a whole.

In **conclusion**, the Science component of the Lakmaga VR platform is a **feasible project with high potential benefits**. It aligns with educational needs, employs suitable technology, and can be realized with the resources and strategies identified. We conclude that the project should move into the implementation phase, with the recommendations provided in this report guiding its execution. With diligent follow-through, this initiative can significantly enhance science education, making learning more interactive, enjoyable, and effective for Sri Lankan students. It transforms the science classroom from a place of rote learning to a space of exploration and discovery – which is ultimately the essence of science itself. The feasibility study thus supports the continuation and expansion of this project, anticipating that it will yield positive outcomes and possibly set a precedent for future educational innovations.

References

1. Vijayatheepan, R. (2025). *Virtual Labs in Science Teaching: Sri Lankan Teachers' Perspectives and Challenges*. Journal of Innovative Practices in Education, 1(2), 76–103. **[18][19]**

2. Lin, X., Li, B., Yao, Z.N., Yang, Z., & Zhang, M. (2024). *The impact of virtual reality on student engagement in the classroom: A critical review of the literature*. *Frontiers in Psychology*, *15*, 1360574. **[4][20]**

3. Labster. (2024). *Inspire Students with Immersive Learning – Labster Virtual Labs are proven to increase STEM course pass rates by building skills and confidence*. [Website]. **[3][17]**

4. National Human Resource Development Council (NHRDC), Sri Lanka. (2024). *Statistical Bulletin on Education – 2023 Edition*. (Reference data: "in the year 2022, around 340,323 candidates sat for the G.C.E. (O/L) examination"). **[2]**

5. Wijenarayana, C.H. (2025). *Gamified VR Educational Platform for Middle School Curriculum in Sri Lanka – Science Module with History: Project Proposal Report*. B.Sc. Thesis, SLIIT. **[1]**

(The above references correspond to sources cited in the feasibility study. Inline citation labels – for example, – refer to specific supporting information from these sources.)

Appendices

Appendix A: Storyboard and Screenshots of Science VR Module – This appendix includes a sample storyboard diagram for the "Pest Control Simulation" scenario, illustrating the sequence of student actions and system responses, as well as annotated screenshots of the virtual paddy field and food chain visualization scenes.

Appendix B: Teacher Training Outline – Detailed outline of the teacher workshop for using the VR module, including agenda (introduction to VR, hands-on practice, integration into lesson planning, safety protocols) and key questions from teachers with provided answers.

Appendix C: Pilot Evaluation Plan – The framework for evaluating the pilot implementation, with sample student and teacher survey questions, and a template for recording observational data during VR sessions (engagement level, technical issues, etc.).